

DK LEARNING

GRADE 12

Advanced Computing & Software Projects

Level: **A/L Advanced** • Purpose: Learn before you practise

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In this guide

1. Algorithm Efficiency
2. Searching and Sorting
3. APIs and Client-Server
4. Abstraction and Modular Design
5. Planning a Software Project
6. Practice questions, answers and glossary



How to use this guide

Read each chapter slowly. Look at the example, then try the “Your turn” task **without copying**. Write your own notes on the lines. When you finish, answer the practice questions and check your work with the answer page. After that, open the Grade 12 activity on the website.

Learning goals

- Compare algorithms using Big-O ideas
- Explain APIs and client-server systems
- Apply abstraction and modular design
- Plan and organise a software project

Study plan

Step	What to do
1	Read the chapter and highlight key words
2	Study the worked example
3	Do the “Your turn” task on paper or on a computer
4	Write a short summary in your own words
5	Answer the practice questions, then check

Tip: When you compare algorithms, think about what happens when the input becomes 1000 times larger.

1 Algorithm Efficiency

Advanced algorithms can be compared by how their resource use grows as input size increases. Big-O notation is a common way to describe this growth. $O(1)$ is constant, $O(n)$ grows in proportion to the input and $O(n^2)$ grows much faster.

Efficiency matters when data is large.

Key points

- $O(1)$ constant
- $O(n)$ linear
- $O(\log n)$ very fast growth control
- $O(n^2)$ slows quickly

Worked example

Searching 1,000,000 items:

Linear search: up to 1,000,000 steps

Binary search (sorted): about 20 steps

Your turn

Explain why binary search is faster but needs sorted data.

Check yourself

Explain each term in your own words:

Big-O

Efficiency

My notes

2 Searching and Sorting

Linear search checks items one by one. Binary search repeatedly halves a sorted list. Sorting arranges items in order. Simple sorts such as bubble sort are easy to understand but slow for large lists.

Choose the method based on data size and whether data is sorted.

Key points

- Linear search: simple
- Binary search: needs sorted data
- Bubble sort: compares neighbours
- Better sorts exist

Worked example

```
def linear_search(items, target):  
    for i, v in enumerate(items):  
        if v == target:  
            return i  
    return -1
```

Your turn

Trace a binary search for 23 in [3, 8, 15, 23, 42].

Check yourself

Explain each term in your own words:

Binary search

Sorting

My notes

3 APIs and Client-Server

An API is an interface that allows software systems to communicate. It defines how requests and responses should be structured. In the client-server model, a client (like a browser) sends a request and a server sends a response.

Data is often sent as JSON.

Key points

- API = agreed way to talk
- Client sends request
- Server sends response
- JSON is a common data format

Worked example

Request: GET /students/12

Response: {"id": 12, "name": "Amal"}

Your turn

Describe a request and response between a weather app and a weather API.

Check yourself

Explain each term in your own words:

API

JSON

My notes

4 Abstraction and Modular Design

Abstraction hides unnecessary implementation detail. Modular design divides a system into components that can be developed and tested separately. Modules should do one job and have clear inputs and outputs.

This makes teamwork and maintenance easier.

Key points

- Hide unneeded detail
- One module, one job
- Clear inputs and outputs
- Easier to test and maintain

Worked example

Student system modules:

Input -> Validation -> Calculation -> Reporting

Your turn

Split a school library system into four modules.

Check yourself

Explain each term in your own words:

Abstraction

Module

My notes

5 Planning a Software Project

A project needs a goal, a list of tasks, a timeline and risks. Break the work into small tasks, order them and assign owners. Review progress often and change the plan when needed.

Document your decisions so the project can be maintained.

Key points

- Goal and scope first
- Break into tasks
- Plan time and risks
- Document decisions

Worked example

Week 1: Requirements
Week 2: Design
Week 3-4: Build
Week 5: Test and fix

Your turn

Create a five-week plan for a student attendance app.

Check yourself

Explain each term in your own words:

Scope

Risk

My notes



Practice Questions

Part A - Choose the best answer

1. Which grows fastest?
 - A) $O(n^2)$
 - B) $O(n)$
 - C) $O(1)$
2. Binary search requires...
 - A) sorted data
 - B) only text
 - C) a network
3. An API allows...
 - A) software to communicate
 - B) printing only
 - C) faster typing
4. Abstraction means...
 - A) hiding unneeded detail
 - B) deleting code
 - C) copying files
5. In client-server, the server...
 - A) sends a response
 - B) types requests
 - C) is always a phone
6. A module should ideally do...
 - A) one job
 - B) every job
 - C) no job

Part B - Short answers

1. What does Big-O describe?

2. Why use modular design?

3. What is JSON?

4. Give one risk in a software project.



Answer Page

Part A

1-A 2-A 3-A 4-A 5-A 6-A

Part B (sample answers)

1. How time or memory grows with input size.
2. Parts can be built, tested and maintained separately.
3. A text format for structured data.
4. Missed deadlines, unclear requirements, bugs, etc.

Mini glossary

Big-O	Notation describing growth of time or memory
Efficiency	How well resources are used
Binary search	Searching a sorted list by halving
Sorting	Putting data in order
API	Application programming interface
JSON	A text format for structured data
Abstraction	Hiding detail to focus on what matters
Module	A self-contained part of a system
Scope	What is and is not included in the project
Risk	Something that could cause problems



Activity Preparation

Design an algorithm, explain a system architecture and implement a small reusable function. Then open the Grade 12 activity.

Think before you start

- Explain one important idea from this grade in your own words.
- Give one real-life example where this topic is useful.
- What mistake could a beginner make, and how can it be prevented?
- Write the steps you will follow before starting the Grade 12 activity.

Ready checklist

- | | |
|---|--|
| ☐ I read every chapter | ☐ I tried the “Your turn” tasks |
| ☐ I understand the key terms | ☐ I checked my practice answers |
| ☐ I wrote my own notes | ☐ I am ready for the activity |

⚡ **START GRADE 12 ACTIVITY**

← **EXIT PDF / BACK TO LEARNING**